

Local Subsidies Fraud Risk - Model Comparison

Note: Korean narrative is in docs/metrics_guide.md. This PDF summarizes aggregate metrics only (no raw rows).

Train period: 202401-202512

Test period: 202507-202512

Risk score range: 0-1000 (higher = higher fraud risk).

Reports contain aggregate statistics only. No raw PII rows.

See docs/metrics_guide.md for Korean explanations of metrics.

Per-algorithm reports: outputs/reports/{algorithm}/

Model metrics (aggregate)

algorithm=catboost_v1,	Accuracy=0.9891878040571551,	Precision=0.22406522350295194,	Recall=0.9224537037037037,
F1=0.36055191133227776,	ROC_AUC=0.9845251209814957		
algorithm=random_forest_v1,	Accuracy=0.9988220175626472,	Precision=0.8232558139534883,	Recall=0.8194444444444444,
F1=0.8213457076566125,	ROC_AUC=0.9828040503929677		
algorithm=catboost_v2,	Accuracy=0.9969364807392223,	Precision=0.5206017004578156,	Recall=0.9212962962962963,
F1=0.6652737150020894,	ROC_AUC=0.9822592095165004		
algorithm=random_forest_v2,	Accuracy=0.9987455251965853,	Precision=0.8109048723897911,	Recall=0.8090277777777778,
F1=0.8099652375434531,	ROC_AUC=0.9853408977857366		